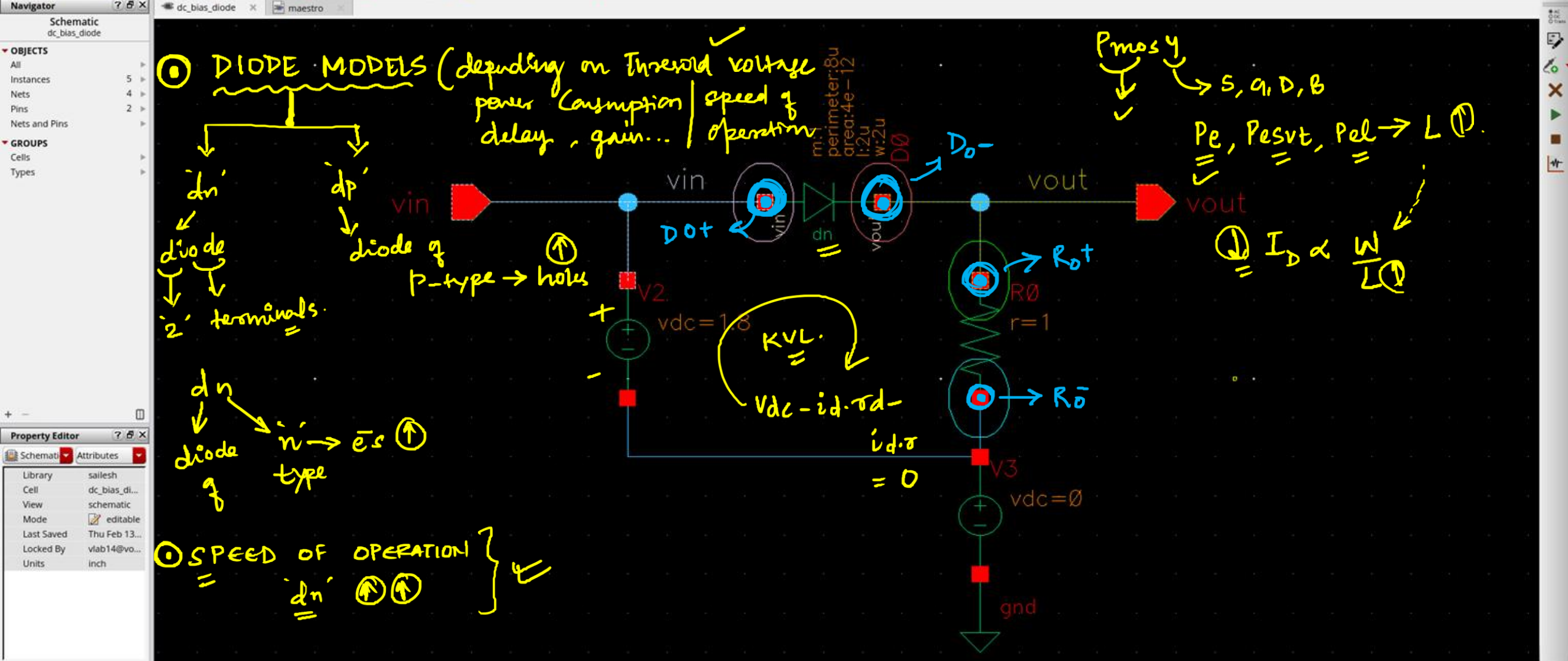


DIODE CHARACTERISTICS

Er. Sailesh Mishra

[Spec. Microelectronics & VLSI]

Silicon University, Bhubaneswar



Filter Filter

sailesh_dc_bias_diode_1

Simulator spectre

Analyses

dc t 0 1.8 Automatic Start-Stop /V2

Click to add analysis

Design Variables

Click to add variable

Parameters

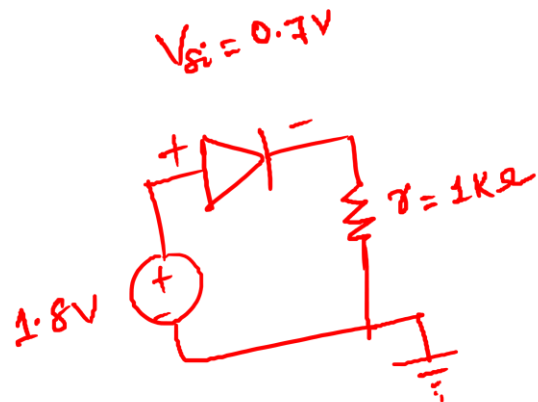
Corners

Reliability Analyses

Monte Carlo Sampling

Checks/Asserts

Name	Type	Details	Value	Plot	Save	Spec
signal (I)	/D0/PLUS			☒	☒	
signal (I)	/D0/MINUS			☒	☒	
signal (I)	/R0/PLUS			☒	☒	
signal (I)	/R0/MINUS			☒	☒	
signal	/vout			☒	☒	
signal	/vin			☒	☒	



Choosing Analyses -- ADE Explorer

Analysis

☒ dc ☐ ac ☐ noise

☐ xf ☐ ens ☐ dcmatch ☐ acmatch

☐ stb ☐ pz ☐ lf ☐ sp

☐ envlp ☐ pss ☐ pac ☐ pstb

☐ pnoise ☐ pxf ☐ psp ☐ qpss

☐ qpac ☐ qpnoise ☐ qpxf ☐ qpssp

☐ hb ☐ hbac ☐ hbstb ☐ hbnoise

☐ hbsp ☐ hbsxf

DC Analysis

Save DC Operating Point ☒

Hysteresis Sweep ☐

Sweep Variable

☐ Temperature ☐ Design Variable ☒ Component Parameter ☐ Model Parameter

Component Name /V2

Select Component

Parameter Name dc

Sweep Range

☒ Start-Stop ☐ Center-Span

Start 0 Stop 1.8

Sweep Type

Automatic

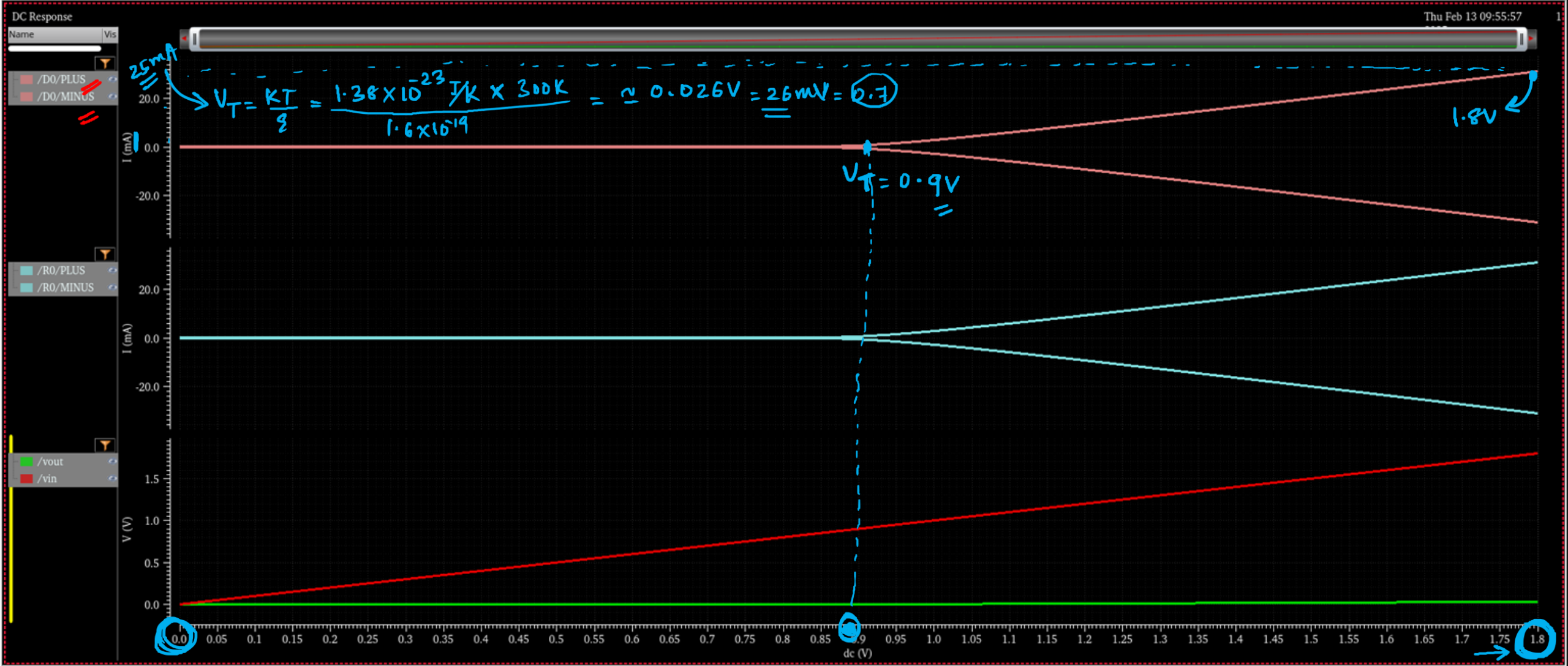
Add Specific Points ☐

Add Points By File ☐

Enabled ☒

Options...

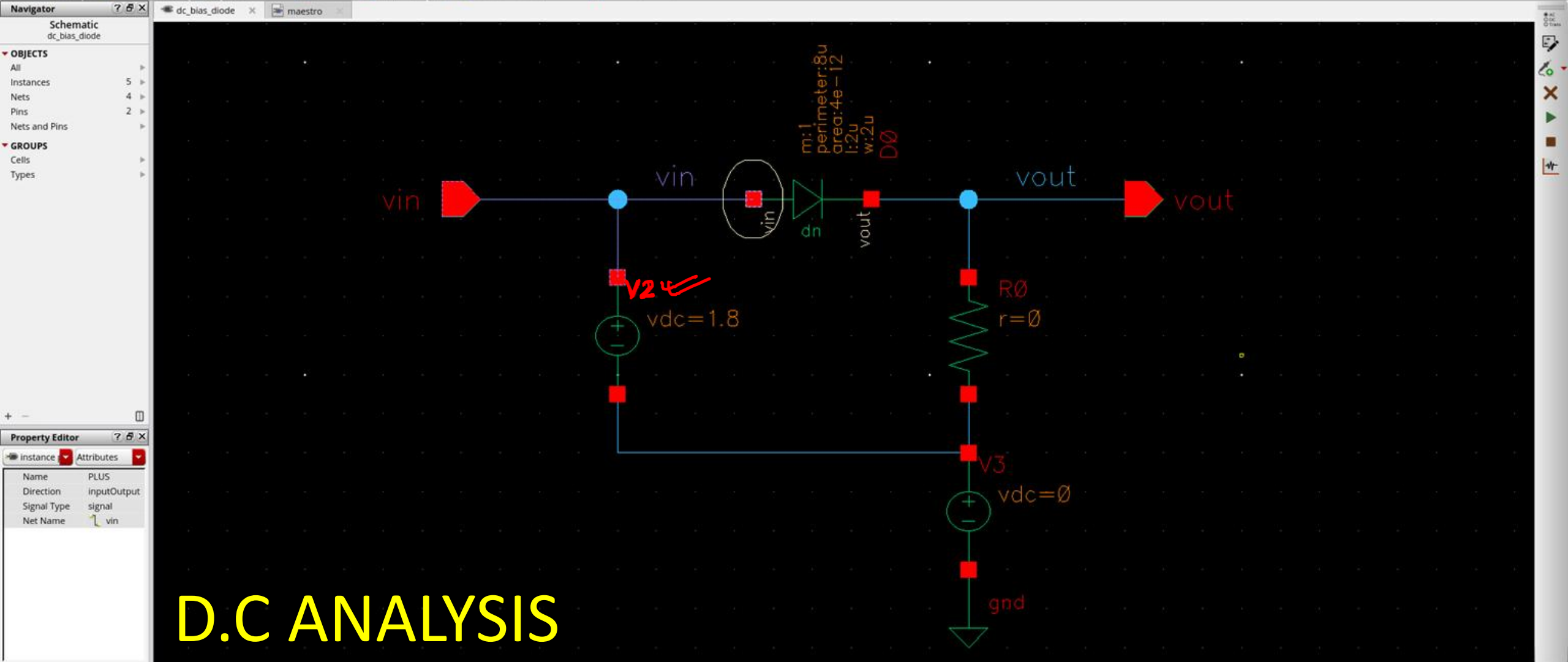
OK Cancel Defaults Apply Help



Launch File Edit View Create Check Options Window Parasitics/LDE ADE Explorer Calibre XKit Utils XT018 Utils PVS Help

Basic Search

Replace (None) dc Sim Time 0



D.C ANALYSIS

Setup

Name Value

Filter Filter

sailesh_dc_bias_diode_1

Simulator spectre

Analyses

dc t 0 1.8 Automatic Start-Stop /V2

Click to add analysis

Design Variables

Click to add variable

Parameters

Corners

Reliability Analyses

Monte Carlo Sampling

Checks/Asserts

Name	Type	Details	Value	Plot	Save	Spec
signal	/vin			☒	☐	
signal (I)	/DO/PLUS			☒	☐	

Choosing Analyses -- ADE Explorer

Analysis

☒ tran ☒ dc ☐ ac ☐ noise

☐ xf ☐ sens ☐ dcmatch ☐ acmatch

☐ stb ☐ pz ☐ lf ☐ sp

☐ envip ☐ pss ☐ pac ☐ pstb

☐ pnoise ☐ pxf ☐ psp ☐ qpss

☐ qpac ☐ qpnoise ☐ qpxf ☐ qpss

☐ hb ☐ hbac ☐ hbstb ☐ hbnoise

☐ hbsp ☐ hbxf

DC Analysis

Save DC Operating Point ☒ → Q-point

Hysteresis Sweep ☐

Sweep Variable

☐ Temperature Component Name /V2

☐ Design Variable Select Component

☒ Component Parameter Parameter Name dc

☐ Model Parameter

Sweep Range

☒ Start-Stop Start 0 Stop 1.8

☐ Center-Span

Sweep Type

Automatic

Add Specific Points ☐

Add Points By File ☐

Enabled ☒

Options...

OK Cancel Defaults Apply Help

Handwritten Notes:

Q-point (Quiescent point)

Still, unchanged, constant.

$y = mx + c$

$\frac{V_{CC}}{R_C}$

BJT → CE → Amplifier

FAITHFUL AMPLIFICATION

o/p characteristics

Q-point $(x, y) = (V_{CE}, I_C)$

CORNER ANALYSIS

BEST CASE SCENARIO

WORST CASE SCENARIO

BGR

Band-gap reference voltage cut

IC

ADC DAC LDO

1.8V

VDD

VDD

VDD

